Lesson 3-5 · Period 3

Real + Complex Zeros, and Modeling

Klimsara-adapted · Student-centered · No Blooket

Do Now — Intersection Points

Algebra 2 · Lesson 3-5 ·

Period 3

DOK 1–2 | 5 min | *bridge from Tuesday*

Practice (3-5-savvas-q11) · Where do the graphs cross? At what points do the graphs of $f(x) = x^3 - 2x^2 - 16x + 20$ and $g(x) = -12$ intersect?

Graph is on your packet. List all intersection points.

Bridge (verbal): A cubic has zeros 2, $1+\sqrt{3}$, and $1-\sqrt{3}$. Are the coefficients rational, irrational, or complex? Why?

Launch — Complex Zeros + Modeling Rules

Algebra 2 · Lesson 3-5 · Period 3

DOK 2 | 12 min | *teacher models*

Complex-Zero Rules (keep on your page)

1. Complex zeros of a real-coefficient poly come in **CONJUGATE PAIRS**.
2. If $a + bi$ is a zero, then $a - bi$ must also be a zero.
3. A degree-n polynomial has exactly n zeros (counting multiplicity).
4. Example: cubic with real zero $x = 4$ and complex zero $2 + 3i \Rightarrow$ third zero is $2 - 3i$.

Explore — TPS + DOK-3 Modeling

Algebra 2 ·

Lesson 3-5 · Period 3

DOK 2–3 | 38 min | *student work*

4 items in order. Plan ∼20 min for Practice #27 (Storage Box).

Try It 3a · Real + complex zeros (cubic) $f(x) = 2x^3 - 8x^2 + 9x - 9$. Find ALL zeros.

Try It 3b · Real + complex zeros (quartic) $f(x) = x^4 - 3x^2 - 4$. Find ALL zeros.

Practice #17 · Real + complex zeros $f(x) = x^3 -$

Share / Summary

DOK 2 **7 min** *DOK-3 reveal + assessment preview*

Sentence frame · Storage Box "For the Storage Box, the factored form was ________. I chose _____ as the length because _________. To find $V = 10\ \text{ft}^3$, I ___________."

Connection to Thursday's Assessment Assessment Q#6 (Lucy's tray) uses the **same move**: factored volume model → extract a hidden dimension.

Today's Storage Box is your rehearsal!

Exit Ticket — Summary Recap

DOK 1–2 | 3 min | *not graded*

Complete on your exit ticket

1. Today I learned that ____________________.
2. For the Storage Box, the hardest step was ____________________.
3. Thursday's assessment will test ____________________.